

Simran Weber

Embedded Systems & IoT Firmware Engineer • 11 Years Experience • MERIQ Index: 70% (Developing)

■ simran.weber993@example.com | ■ +1 (548) 282-8530 | ■ Berlin, Germany

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: ARM Cortex, FreeRTOS, I2C/SPI/UART, Python, C/C++, PCB Debugging, Embedded Linux

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core ARM Cortex service, reducing latency by 37%.
- Led a squad of 4 engineers delivering scalable microservices processing 7M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 42%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2015 - 2023)

- Designed and maintained FreeRTOS data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using I2C/SPI/UART saving \$25k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [ARM Cortex, FreeRTOS, I2C/SPI/UART, Python]
Engineered end-to-end production solution leveraging ARM Cortex and FreeRTOS with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [I2C/SPI/UART, Python, C/C++, PCB Debugging]

Architected high-resilience system integrating I2C/SPI/UART with 99.9% uptime.

EDUCATION & CREDENTIALS

M.Tech in Data Science & Artificial Intelligence

IIT Delhi — 2011 - 2015 (Score: 8.91 / 10.0)